Chang **Liu**

TIME-DOMAIN ASTRONOMER

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Education

PhD Northwestern University — *Evanston, US* *Jun 2026*
MSc Northwestern University — *Evanston, US* *Jun 2023*
BSc (Hons) Peking University — *Beijing, China* *Jun 2020*

Research Interests

- **Survey Science & Machine Learning:** Leveraging advanced statistical and machine learning techniques to identify population-level patterns of transients in the Rubin/LSST era.
- **Supernovae (SNe):** Infant SNe, nebular-phase spectroscopy, spectral modeling.
- **Tidal disruption events (TDEs):** Repeating partial TDEs, exotic nuclear transients.
- **Software:** Open source software development, data reduction pipelines.

Skills

Astronomy Transient follow-up, spectroscopic observations and data reduction, hydrodynamical simulations
Data Science Hierarchical Bayesian modeling, Gaussian processes, gradient boosting
Programming **Proficient:** Python **Experienced:** Shell, Fortran, SQL
Softwares
- HostSub_GP (**developer**): removing host galaxy background in transient long-slit spectroscopy using Gaussian processes and JAX accelration
- BayeSpecFit (**developer**): fitting blended SN spectral features with Bayesian inference
- PypeIt: optical/NIR spectrum reduction

Telescope Experience

Obs Runs Keck I (LRIS ‖ 6.5 n, MOSFIRE ‖ 0.5 n), Keck II (DEIMOS ‖ 1.5 n, KCWI ‖ 0.5 n) *2023 – 2026*
Obs Runs Magellan Baade (FIRE ‖ 3 n) *2025 – 2026*
PI 2.56 m Nordic Optical Telescope (NOT), NOIRLab ‖ 7 hr *2026B*
PI 4.1 m Southern Astrophysical Research (SOAR) Telescope, NOIRLab ‖ 7 hr *2026B*
PI 6.5 m Magellan Baade Telescope, Northwestern ‖ 1 n *2026A*
PI 2.56 m Nordic Optical Telescope (NOT), NOIRLab ‖ 8 hr *2026A*
PI 10 m Keck Telescopes, Northwestern ‖ 1.5 n *2025B*
PI 6.5 m Magellan Baade Telescope, Northwestern ‖ 1 n *2025B*
PI 2.56 m Nordic Optical Telescope (NOT), NOIRLab ‖ 8 hr *2025B*
PI 6.5 m Magellan Baade Telescope, Northwestern ‖ 1 n *2025A*
PI 2.56 m Nordic Optical Telescope (NOT), NOIRLab ‖ 6 hr *2025A*
PI 4.1 m Southern Astrophysical Research (SOAR) Telescope, NOIRLab ‖ 10 hr *2025A*
PI 10 m Keck Telescopes, Northwestern ‖ 2 n *2024B*
co-I James Webb Space Telescope, NIRSpec + NIRCam (PI Goobar) ‖ 10.2 hr *DD 12764, Cycle 4*
co-I James Webb Space Telescope, NIRSpec + MIRI (PI Jacobson–Galán) ‖ 6 hr *DD 9492, Cycle 4*
co-I James Webb Space Telescope, NIRSpec + MIRI (PI Kwok) ‖ 12.9 hr *DD 6811, Cycle 3*

Invited Talks & Seminars

Seminar	Carnegie arXiv Tea, Carnegie Observatories	*Pasadena, US*	*Sep 2025*
Seminar	Astronomy Tea Talk, California Institute of Technology	*Pasadena, US*	*Sep 2025*
Seminar	DESI Special Seminar	*Berkeley, US*	*Jun 2025*
Seminar	University of California, Santa Cruz	*Santa Cruz, US*	*Jun 2025*
Invited Talk	LS4 Team Meeting	*Evanston, US*	*Mar 2025*
Seminar	Peking University	*Beijing, China*	*Dec 2023*
Seminar	Shanghai Jiao Tong University	*Shanghai, China*	*Jun 2023*
Seminar	Astro-Coffee@Princeton	*Princeton, US*	*Sep 2022*

Contributed Talks & Posters

Talk	MMT Science Symposium	*Tucson, US*	*Apr 2026*
Talk	247th AAS Meeting	*Phoenix, US*	*Jan 2026*
Talk	Keck Science Meeting	*Los Angeles, US*	*Sep 2025*
Poster	Open SkAI 2025	*Chicago, US*	*Sep 2025*
Poster	CIERA Fellows at 15	*Evanston, US*	*Aug 2025*
Talk	One Hundred Years of Supernova Science	*Saltsjöbaden, Sweden*	*Aug 2025*
Talk	Cosmic Lighthouses: Astrophysical and Cosmological Challenges with Type Ia Supernovae	*Cambridge, UK*	*Jul 2025*
Poster	Center for Decoding the Universe Annual Conference	*Stanford, US*	*Jun 2025*
Poster	CoDEx: Symposium for Computation and Data Intensive Research	*Evanston, US*	*Apr 2025*
Poster	Transients from Space	*Baltimore, US*	*Mar 2025*
Poster	Rise_Time 2024	*West Lafayette, US*	*Aug 2024*
Talk	243rd AAS Meeting	*New Orleans, US*	*Jan 2024*
Talk	The 32nd Texas Symposium on Relativistic Astrophysics	*Shanghai, China*	*Dec 2023*
Talk	ZTF Science Meeting	*Pasadena, US*	*Oct 2023*
Poster	Keck Science Meeting	*Berkeley, US*	*Sep 2023*
Talk	SPOKEN-WERRD 2022 Symposium	*Virtual*	*Nov 2022*
Talk	ZTF Science Meeting	*Evanston, US*	*Oct 2022*
Talk	PKU-DoA Undergraduate Astronomy Symposium	*Beijing, China*	*Sep 2020*

Honors & Awards

Oct 2026	**Hintze Prize Fellowship**, Hintze Centre for Astrophysical Surveys, the University of Oxford
Oct 2020	**First prize & Lin-bridge Prize for Excellent Undergraduate Research**, 2020 PKU-DoA Undergraduate Astronomy Symposium (endowed by Prof. Douglas Lin)
May 2020	**Outstanding Graduate**, College Graduate Excellence Award of Beijing
May 2020	**Outstanding Graduate**, College Graduate Excellence Award of Peking University
Jun 2019	**Caltech Summer Undergraduate Research Fellowship (SURF)**, California Institute of Technology
May 2019	**National Innovation Training Program**, Undergraduate Research & Training Program
Oct 2018	**Wei Lin Scholarship** , Annual scholarship of 2017-2018, Peking University
Oct 2017	**Merit Student Pacesetter (the Utmost Annual Accolade at PKU)**, Annual honor of 2016-2017, Peking University
Oct 2017	**Arawana Scholarship**, Annual scholarship of 2016-2017, Peking University

Professional Service

Referee	Astrophysical Journal
Referee	Astrophysical Journal Letters
Referee	Science Bulletin

Outreach

Lecturer *Evanston, US*
Research Experience in Astronomy at CIERA for High School Students (REACH) *Jul 2023*

Invited Speaker (*supernovae: from the past to the future*) *Beijing, China*
Peking University Youth Astronomy Society (PKU-YAS) *Apr 2021*

Volunteer *Beijing, China*
Summer Camp of Astronomy for High School Students, Peking University *Jul 2018*

References

Prof. Adam A. Miller *Evanston, US*
amiller@northwestern.edu *Department of Physics and Astronomy, Northwestern University*

Prof. Enrico Ramirez-Ruiz *Santa Cruz, US*
enrico@ucolick.org *Department of Astronomy and Astrophysics, UC Santa Cruz*

Prof. Xian Chen *Beijing, China*
xian.chen@pku.edu.cn *Department of Astronomy, Peking University*

Publication List

10 first-author papers (including **1** co-first authorship) among **39** total publications (h-index: **15**)

First-author Publications

- **C. Liu** & A. A. Miller., Decoding the Early-Time Light Curves of Type Ia Supernovae. I. A Hierarchical Bayesian Framework for Demographic Inference, *arXiv: 2607.00075, submitted to ApJ.*
- **C. Liu**, A. A. Miller, N. Sarin, et al., Decoding the Early-Time Light Curves of Type Ia Supernovae. II. Population Parameters of One Thousand ZTF Supernovae, *arXiv: 2607.00081, submitted to ApJ.*
- **C. Liu**, **J. Hinkle**, A. A. Miller, et al., A Disappearing Act: Constraints From "Missing" Flares of Repeating Partial TDE Candidates, *arXiv: 2606.06578, submitted to ApJL.*
- **C. Liu** & A. A. Miller., HostSub_GP: Precise Galaxy Background Subtraction in Transient Spectroscopy with Gaussian Processes, *2026, PASP, 138, 024503.*
- **C. Liu**, A. A. Miller, J. S. Bloom, et al., A Morphological Model to Separate Resolved–Unresolved Sources in the DESI Legacy Surveys: Application in the LS4 Alert Stream, *2025, PASP, 137, 084501.*
- **C. Liu**, R. Yarza, & E. Ramirez-Ruiz., Repeating Partial Tidal Encounters of Sun-like Stars Leading to their Complete Disruption, *2025, ApJ, 979, 40.*
- **C. Liu**, A. A. Miller, S. J. Boos, et al., SN 2022joj: A Peculiar Type Ia Supernova Possibly Driven by an Asymmetric Helium-shell Double Detonation, *2023, ApJ, 958, 178.*
- **C. Liu**, A. A. Miller, A. Polin, et al., SN 2020jgb: A Peculiar Type Ia Supernova Triggered by a Helium-Shell Detonation in a Star-Forming Galaxy, *2023, ApJ, 946, 83.*
- **C. Liu**, B. Mockler, E. Ramirez-Ruiz, et al., Tidal Disruption Events from Eccentric Orbits and Lessons Learned from the Noteworthy ASASSN-14ko, *2023, ApJ, 944, 184.*
- **C. Liu**, X. Chen, & F. Du, Impact of an Active Sgr A* on the Synthesis of Water and Organic Molecules Throughout the Milky Way, *2020, ApJ, 899, 2.*

Co-author Publications

- S. Belkin et al. (incl. **C. Liu**), GOTO identification and broadband modelling of the counterpart to the SVOM GRB 250818B, *2026, MNRAS, 550, 2.*
- T. E. Müller-Bravo et al. (incl. **C. Liu**), ZTF SN Ia DR2 follow-up: early excess in Type Ia supernova light curves, *2026, A&A in press.*
- S. Hall et al. (incl. **C. Liu**), JWST Observations of Calcium-Strong Transients: I. Complex Nebular He Emission in SN 2024uj, *arXiv: 2607.00111, submitted to ApJ.*
- J. H. Terwel et al. (incl. **C. Liu**), An analysis of the Type Ia SN 2024gy and a comparison of different host extinction estimation techniques, *2026, A&A in press.*

- D. A. Perley et al. (incl. **C. Liu**), AT2024wpp: An Extremely Luminous Fast Ultraviolet Transient Powered by Accretion onto a Black Hole, *2026, MNRAS, 549, 1.*
- P. Chen et al. (incl. **C. Liu**), Detection of persistent helium absorption in the 91bg-like type Ia Supernova 2022an, *2026, ApJL, 1004, L42.*
- C. L. Ransome et al. (incl. **C. Liu**), The Eye of Sauron in SN 2025ngs: a Short-plateau Cousin of SN 1998S with Evidence for a Ring-like Circumstellar Medium, *2026, ApJ in press.*
- R. Hwangbo et al. (incl. **C. Liu**), Near-Infrared and Optical Observations of SN 2024rbc: The First Early Detection of CO and Dust in a Type Ib Supernova, *arXiv: 2603.23877, submitted to ApJ.*
- A. Singh et al. (incl. **C. Liu**), Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link Among Type Iax Supernovae, *2026, ApJ, 999, 227.*
- C. Sevilla et al. (incl. **C. Liu**), Multiwavelength Analysis of Six Luminous, Fast Blue Optical Transients, *2026, ApJ in press.*
- M. M. Kasliwal et al. (incl. **C. Liu**), ZTF25abjmnps (AT2025ulz) and S250818k: A Candidate Superkilonova from a Sub-threshold Sub-Solar Gravitational Wave Trigger, *2025, ApJL, 995, L59.*
- J. Pearson et al. (incl. **C. Liu**), Mid-Infrared Dust Evolution and Late-time Circumstellar Medium Interaction in SN 2017eaw, *2025, ApJ, 993, 213.*
- A. C. Gordon et al. (incl. **C. Liu**), Mapping the Spatial Distribution of Fast Radio Bursts within their Host Galaxies, *2025, ApJ, 993, 119.*
- A. A. Miller et al. (incl. **C. Liu**), The La Silla Schmidt Southern Survey, *2025, PASP, 137, 094204.*
- L. A. Kwok, **C. Liu**, S. W. Jha, et al., JWST Spectroscopy of SN Ia 2022aaiq and 2024gy: Evidence for Enhanced Central Stable Ni Abundance and a Deflagration-to-Detonation Transition, *2026, ApJ, 1005, 52.*
- P. J. Pessi et al. (incl. **C. Liu**), The ambiguous AT2022rze: Changing-look AGN mimicking a supernova in a merging galaxy system, *2025, MNRAS, 542, 4.*
- L. A. Kwok et al. (incl. **C. Liu**), JWST and Ground-based Observations of the Type Iax Supernovae SN 2024pxl and SN 2024vjm: Evidence for Weak Deflagration Explosions, *2025, ApJL, 989, L33.*
- A. Y. Q. Ho et al. (incl. **C. Liu**), A Luminous Red Optical Flare and Hard X-ray Emission in the Tidal Disruption Event AT2024kmq, *2025, ApJ, 989, 54.*
- J. C. Rastinejad et al. (incl. **C. Liu**), EP 250108a/SN 2025kg: Observations of the most nearby Broad-Line Type Ic Supernova following an Einstein Probe Fast X-ray Transient, *2025, ApJL, 988, L13.*
- Y. Yao et al. (incl. **C. Liu**), A Massive Black Hole 0.8 kpc from the Host Nucleus Revealed by the Offset Tidal Disruption Event AT2024tvd, *2025, ApJL, 985, L48.*
- N. Rehemtulla et al. (incl. **C. Liu**), The BTSbot-nearby discovery of SN 2024jlf: rapid, autonomous follow-up probes interaction in an 18.5 Mpc Type IIP supernova, *2025, ApJ, 985, 241.*
- K. Das et al. (incl. **C. Liu**), Low-Luminosity Type IIP Supernovae from the Zwicky Transient Facility Census of the Local Universe. I: Luminosity Function, Volumetric Rate, *2025, PASP, 137, 044203.*
- L. Harvey et al. (incl. **C. Liu**), ZTF SN Ia DR2: High-velocity components in the Si IIλ6355, *2025, A&A, 695, A264.*
- T. Eftekhari et al. (incl. **C. Liu**), The Massive and Quiescent Elliptical Host Galaxy of the Repeating Fast Radio Burst FRB 20240209A, *2025, ApJL, 979, L22.*
- G. Dimitriadis et al. (incl. **C. Liu**), ZTF SN Ia DR2: The diversity and relative rates of the thermonuclear supernova population, *2024, A&A, 694, A10.*
- Z. Wu et al. (incl. **C. Liu**), Gaia22dkvLb: A Microlensing Planet Potentially Accessible to Radial-Velocity Characterization, *2024, AJ, 168, 62.*
- K. Das et al. (incl. **C. Liu**), SN 2023zaw: an ultra-stripped, nickel-poor supernova from a low-mass progenitor, *2024, ApJL, 969, L11.*
- P. Chen et al. (incl. **C. Liu**), A 12.4 Day Periodicity in a Close Binary System after a Supernova, *2023, Nature, 625, 7994, 253-258.*
- G. Dimitriadis et al. (incl. **C. Liu**), SN 2021zny: an early flux excess combined with late-time oxygen emission suggests a double white dwarf merger event, *2023, MNRAS, 521, 1162.*